Warmup/Attendance: [Week 3: Friday 9/11/2026]

Full Name: _____ SID: _____

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. What will this line of code return?

```
c("smile", "frown", "smile") == "frown"
```

2. Rewrite this nested function call using pipes

```
exp(mean(c(1, 2, 3, NA), na.rm = TRUE))
```

3. Filtering

```
flights |>
  filter(dep_time >= 22,
         carrier %in% c("UA", "AA"),
         dep_delay <= 0)
```

Which observations will be included in the data frame produced by this code? Or equivalently, which observations will *not* be included?

Just write a few English sentences describing what flights will remain in the new dataframe…

4. Which data frame will have fewer rows?

```
# A
filter(flights, origin == "OAK")

# B
flights |>
  mutate(from_oakland = (origin == "OAK")) |>
  filter(from_oakland)
```

5. Build a pipeline

Write a code pipeline that extracts, from the penguins dataframe, only Adelie penguins with bill lengths between 40 and 45mm.

6. What does this line of code compute?

```
penguins |>
  mutate(chonky = body_mass_g > 3500) |>
  summarize(mean(chonky))
```

Answer in an English sentence. Nevermind what actual number it will produce, just explain what that number will mean.